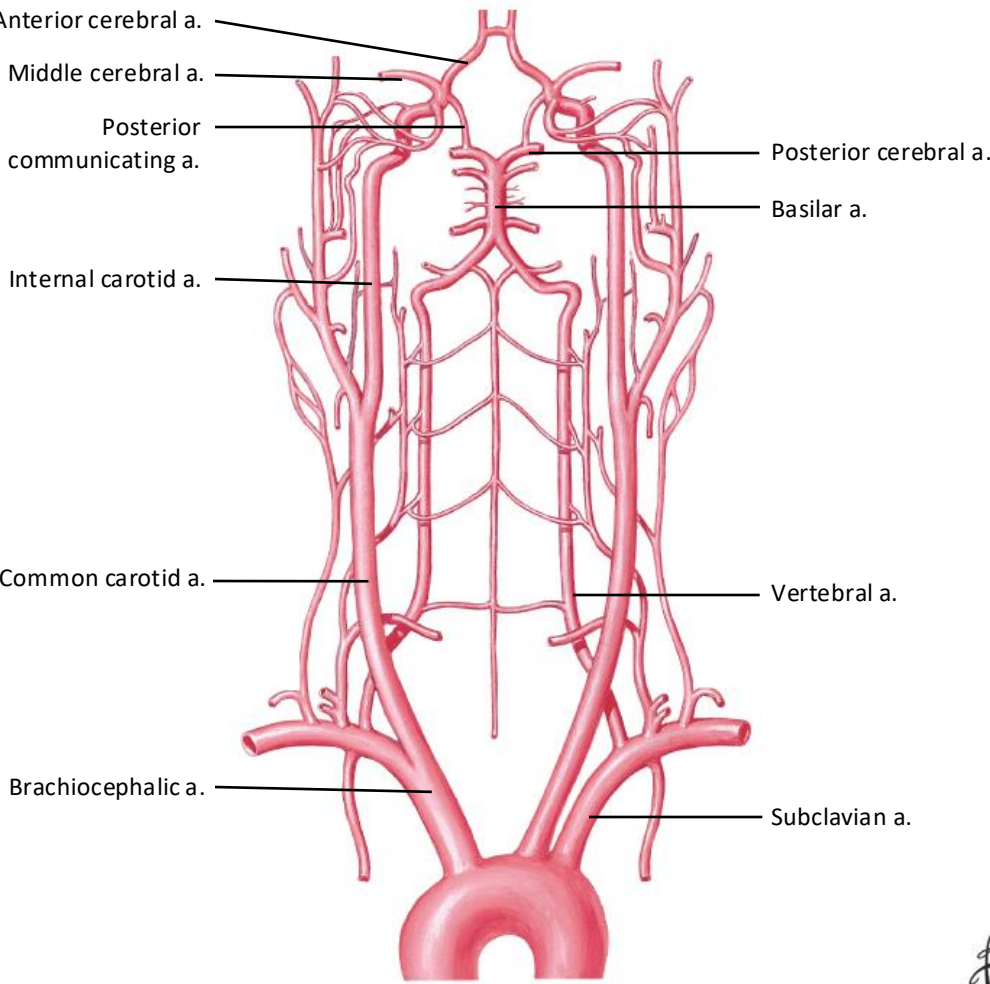


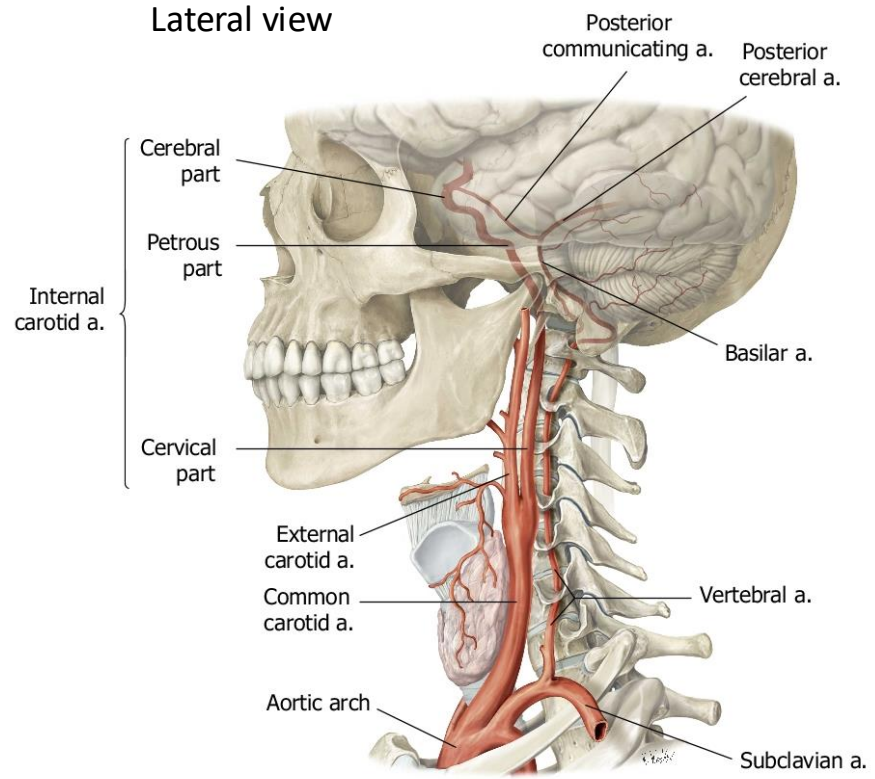
Anterior supply

Arteries to Brain: Schema

Posterior supply



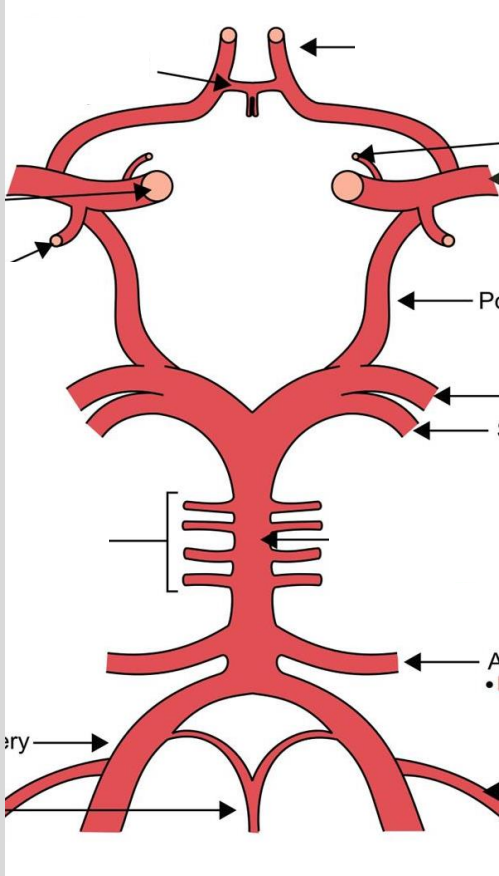
Lateral view



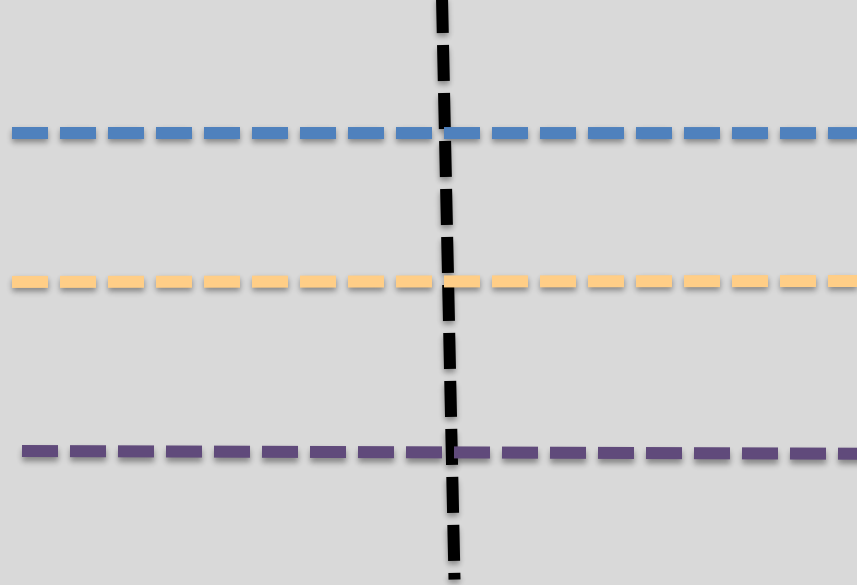
Where would you position these lines?

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Midline



Forebrain

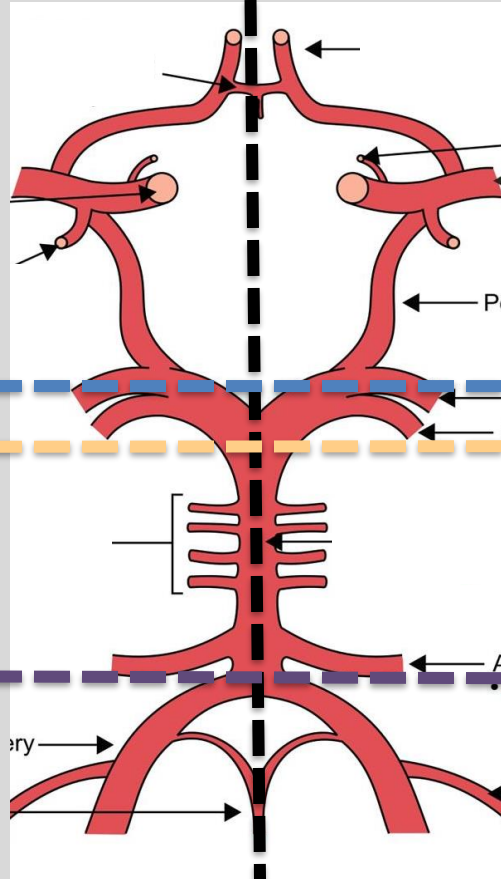
Midbrain

Pons

Medulla

Pause here.....

Midline



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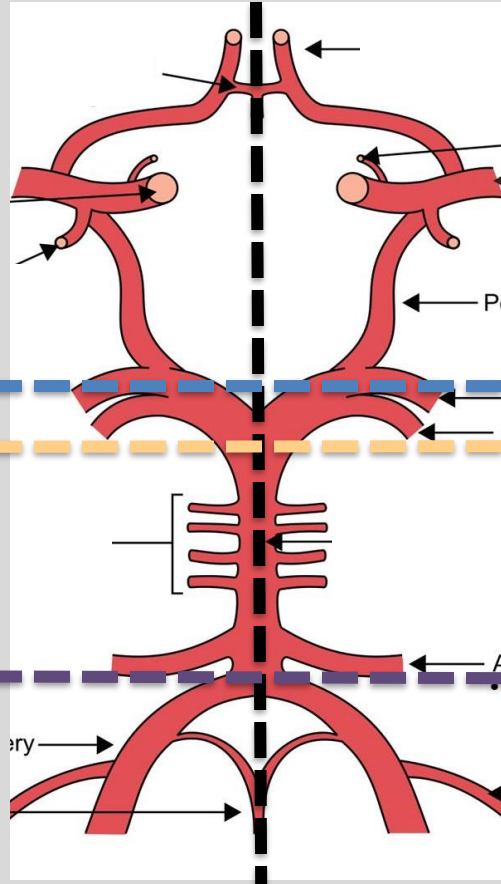
Forebrain

Midbrain

Pons

Medulla

Midline



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Forebrain

Midbrain

Pons

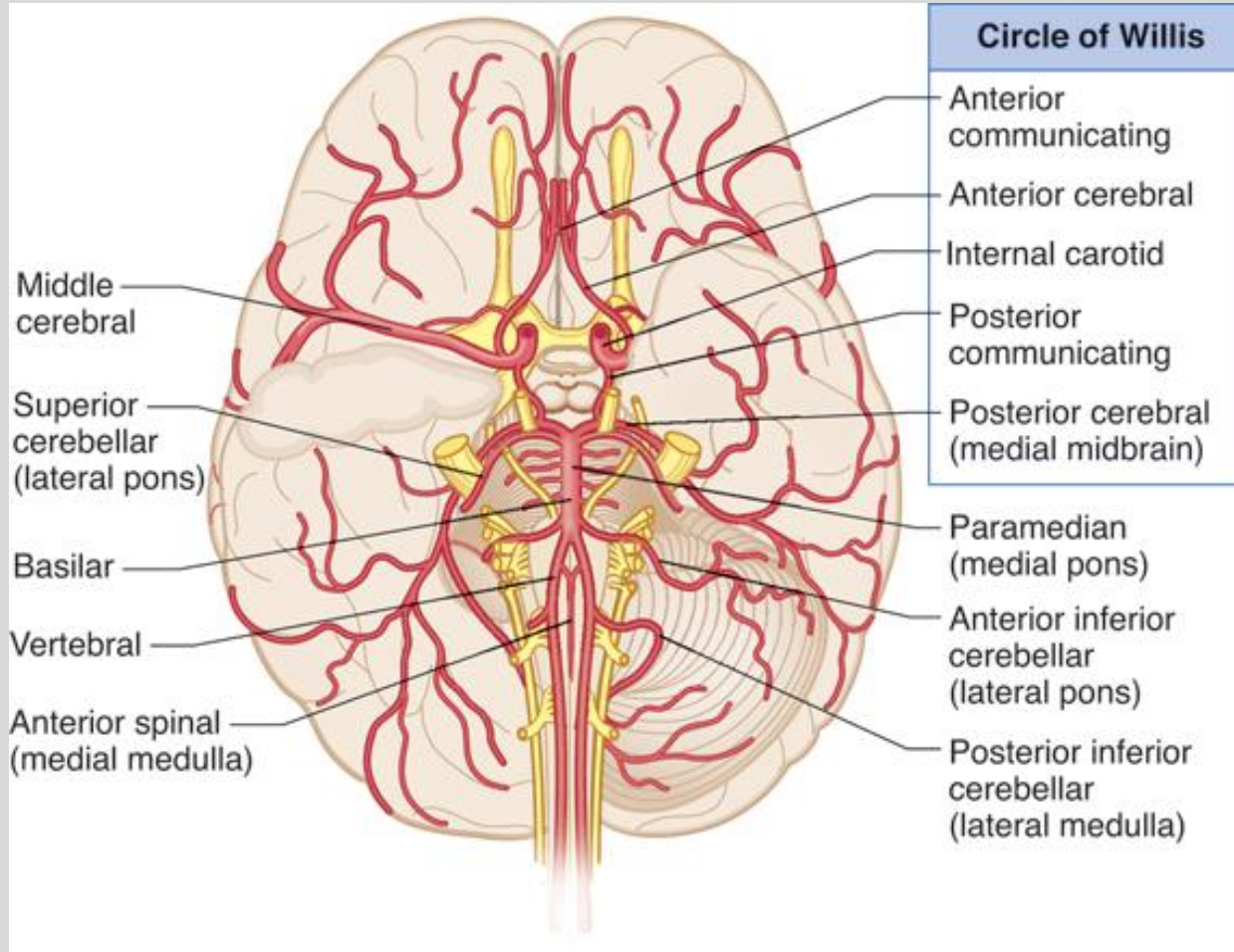
Medulla

1
2
3
4
5
6
7
8
9
10
11
12

Pause here.....

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Major Structures & Functions

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1)

2)

3)

4)

5)

6)

7)

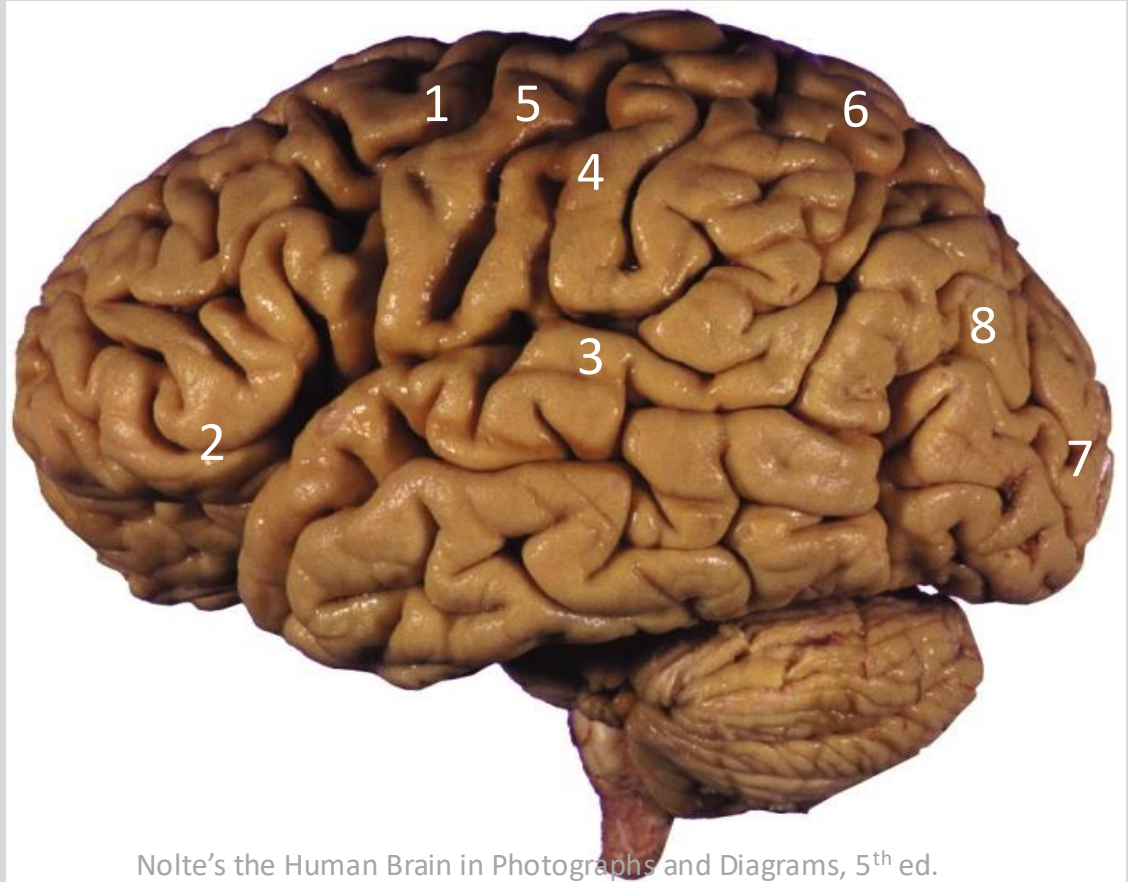
8)



Major Structures & Functions

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- 1) Pre-motor
 - Planning movements
 - Frontal eye field
- 2) Primary gustatory
(anterior insula & frontal operculum)
- 3) Primary Auditory - Both ears
- 4) Primary Somatosensory
- 5) Primary Motor
- 6) Somatosensory Association
- 7) Primary Visual
- 8) Visual Association
 - Motion & Location



Nolte's the Human Brain in Photographs and Diagrams, 5th ed.

Major Structures & Functions

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1)

2)

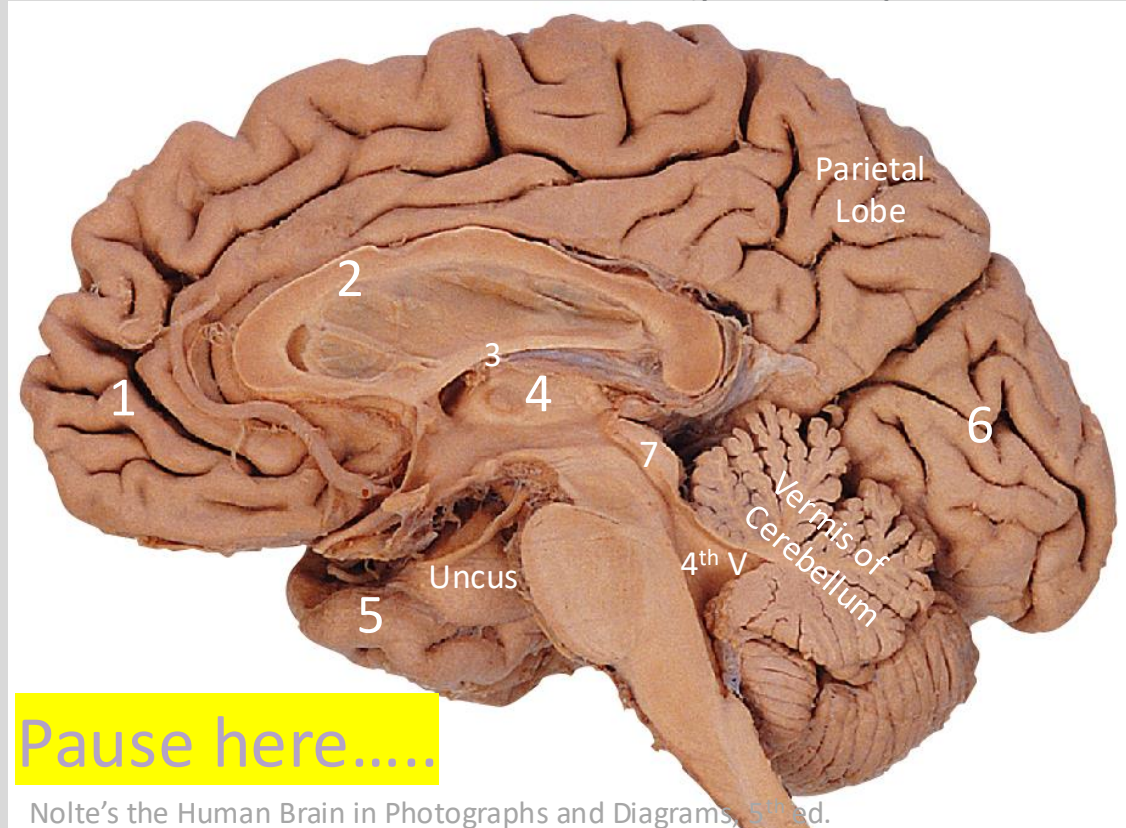
3)

4)

5)

6)

7)



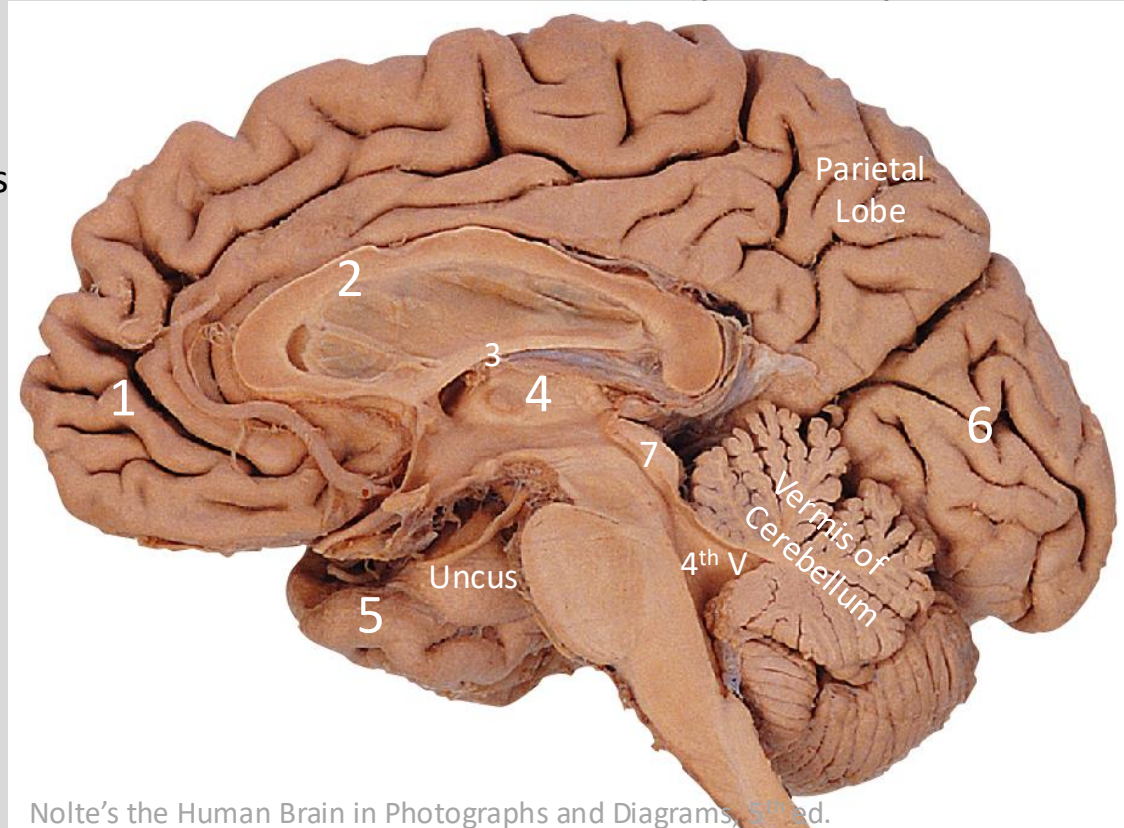
Nolte's the Human Brain in Photographs and Diagrams, 5th ed.

Major Structures & Functions

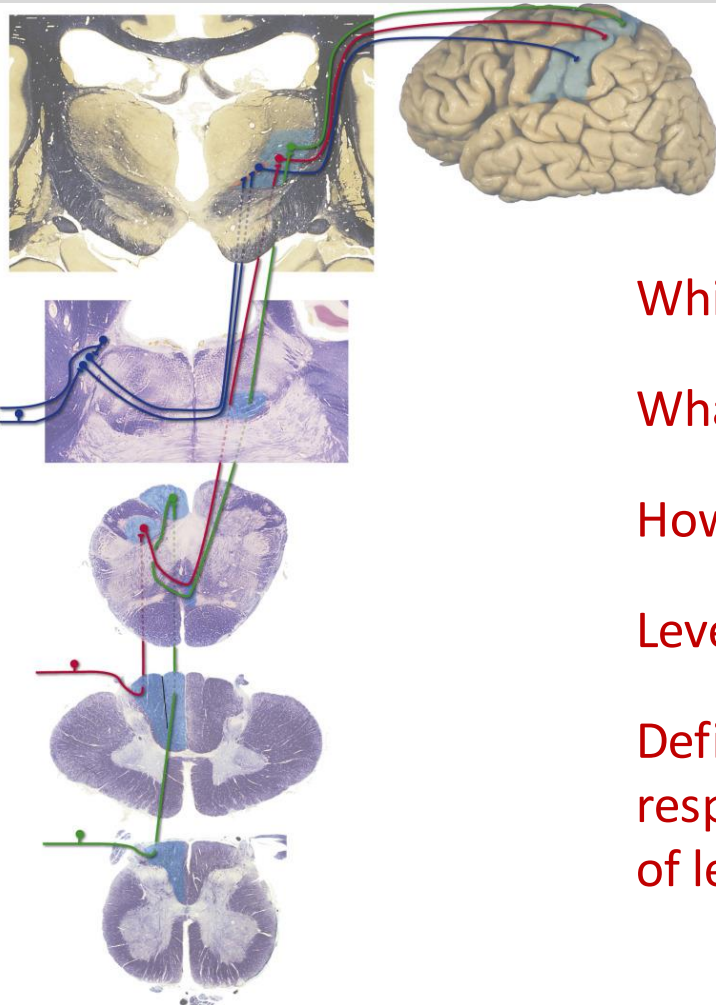
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- 1) Pre-frontal cortex
 - Personality
 - Executive functions
- 2) Corpus callosum
 - Cerebral hemisphere communications
- 3) Fornix
 - Output of the hippocampus
- 4) Thalamus
 - Sensory relay & association
- 5) Piriform cortex
 - Primary olfactory cortex
- 6) Calcarine sulcus
 - Primary visual cortex
- 7) Tectum
 - Superior colliculus -vision
 - Inferior colliculus -auditory



Nolte's the Human Brain in Photographs and Diagrams, 5th ed.



Which tract?

What modality?

How many neurons relay?

Level of decussation?

Deficits lateralization in
respect to different levels
of lesions?

Pause here.....

Ascending

Dorsal Column – Medial Lemniscus (DCML) Pathway

Which tract? - DCML

What modality?

– Sensory, touch/proprioception

How many neurons relay?

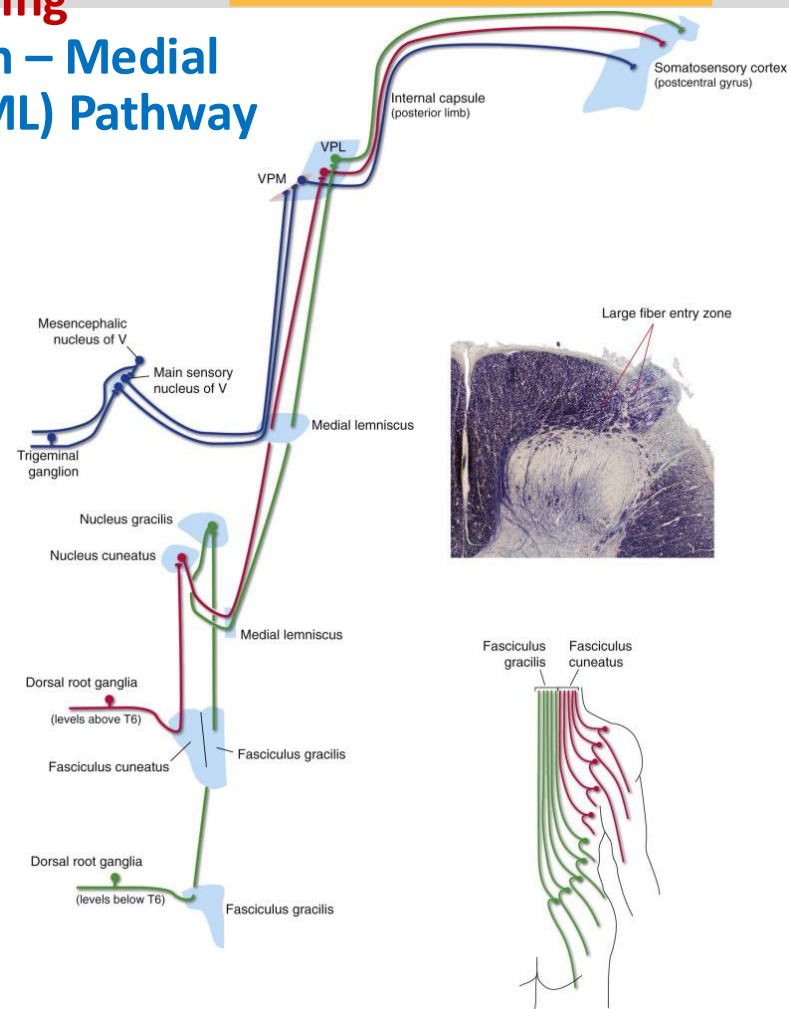
– 3 neurons: 1st order (primary afferents, DRG), 2nd order (medulla), 3rd order (VPL)

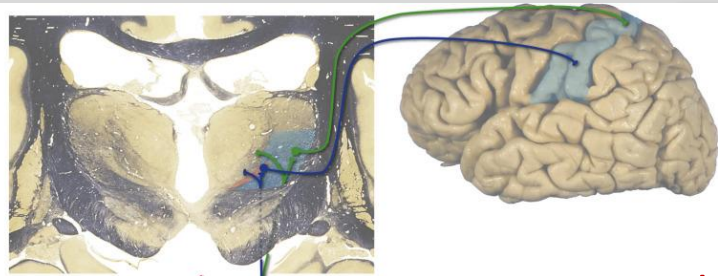
Level of decussation?

– Caudal medulla (n. gracilis & cuneatus)

Deficits lateralization?

– Contralateral to the lesion when above medulla; ipsilateral to the lesion when below medulla (spinal cord)





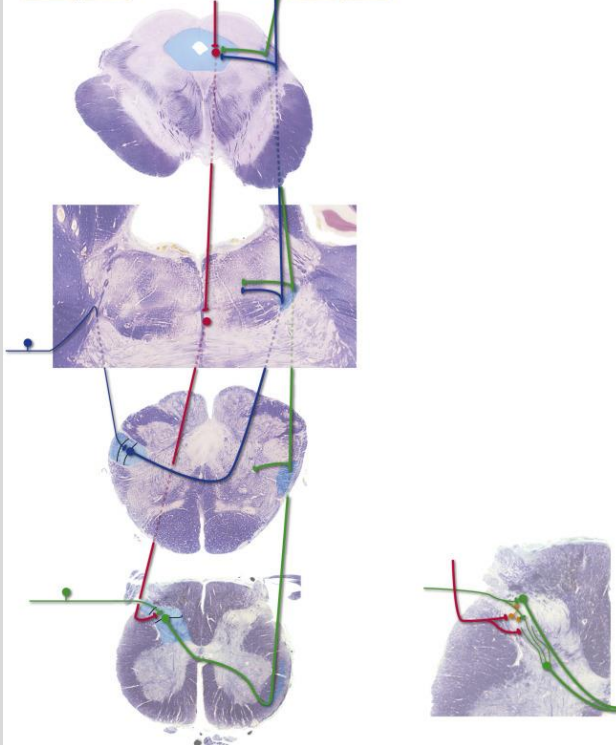
Which tract?

What modality?

How many neurons relay?

Level of decussation?

Deficits lateralization?



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Pause here.....

Ascending Spinothalamic Tract

Which tract? - STT

What modality?

– Sensory, pain/temperature

How many neurons relay?

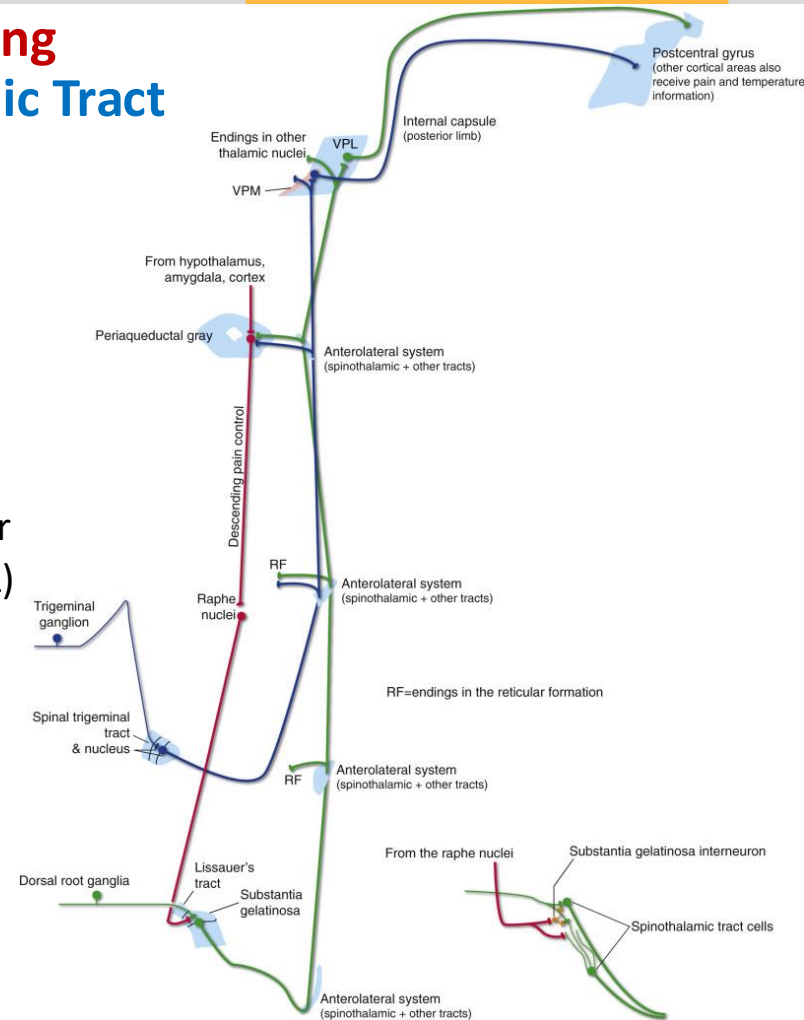
– 3 neurons: 1st order (DRG), 2nd order (dorsal horn), 3rd order (thalamus VPL)

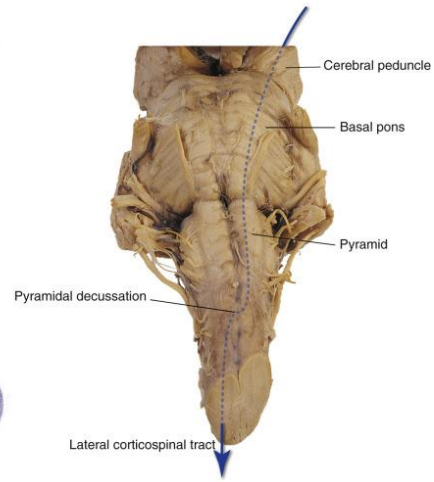
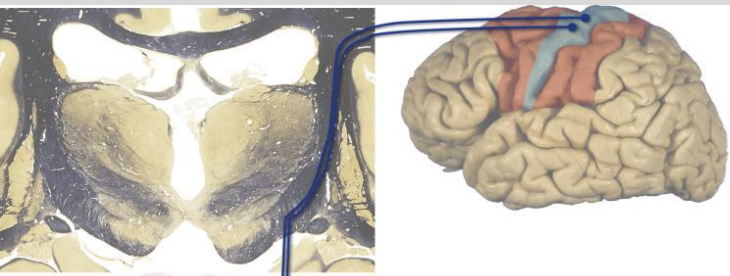
Level of decussation?

– Spinal cord, 1-2 segments above the primary afferents level

Deficits lateralization?

– Almost always contralateral to the lesion, except for primary afferents or dorsal horn lesions





Which tract?

What modality?

How many neurons relay?

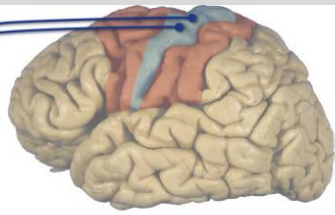
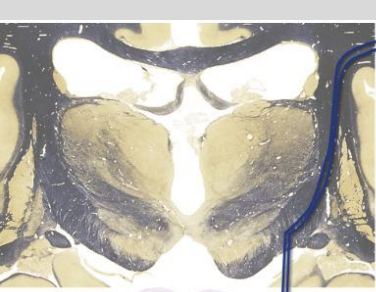
Level of decussation?

Deficits lateralization?

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Pause here.....



Descending Corticospinal Tract (pyramidal tract)

Which tract? - CST

What modality?

– Motor, voluntary movement

How many neurons relay?

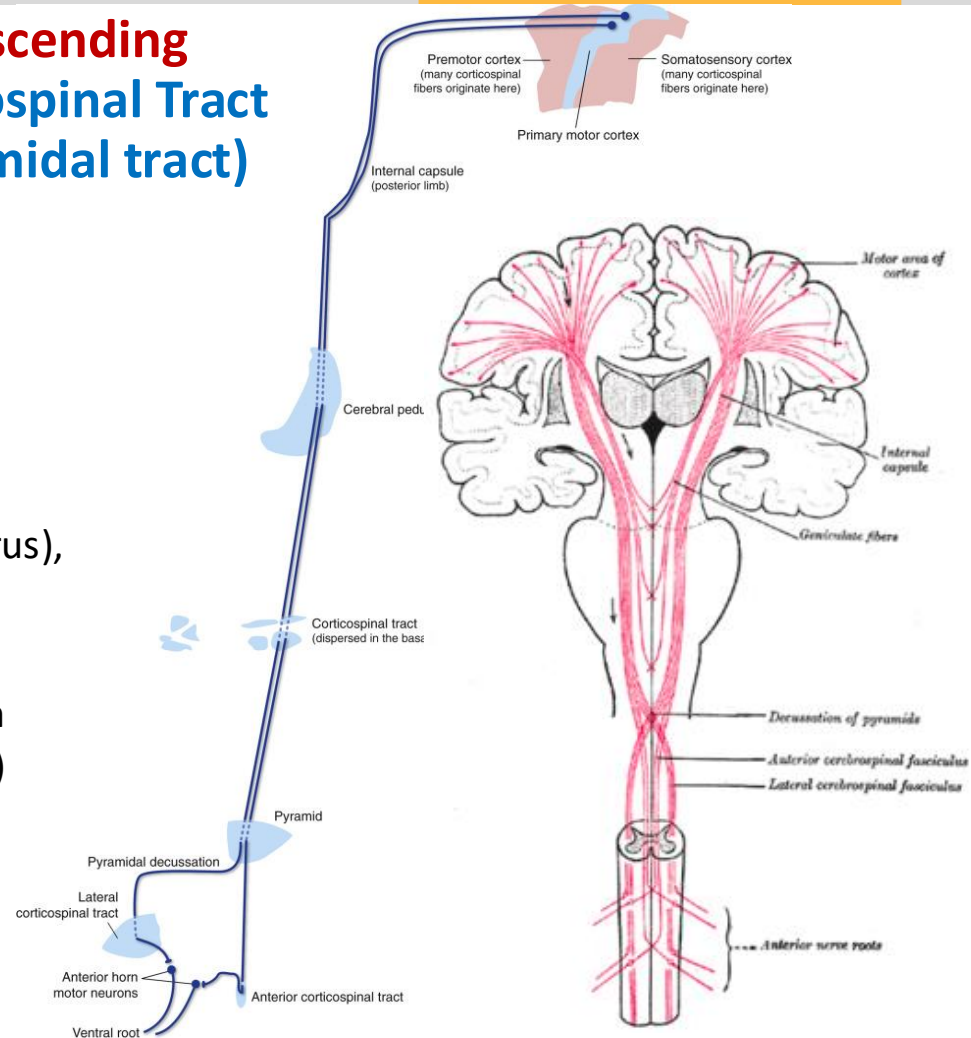
– 2 neurons: UMN (precentral gyrus),
LMN (ventral horn)

Level of decussation?

– Pyramidal decussation (junction
between medulla and spinal cord)

Deficits lateralization?

– Contralateral to the lesion
above the spinal cord; ipsilateral
to the lesion at the spinal cord



Lesions

		UMN	LMN
Cortico-spinal tract	Contraction	↑	↓
	Reflex	↑	↓
	Babinski sign	Y	N
	Lateralization	ipsi/ contra	ipsi

Cortico-bulbar tract	Contraction	None	↓
	Reflex	None	↓
	Lateralization	NA	ipsi

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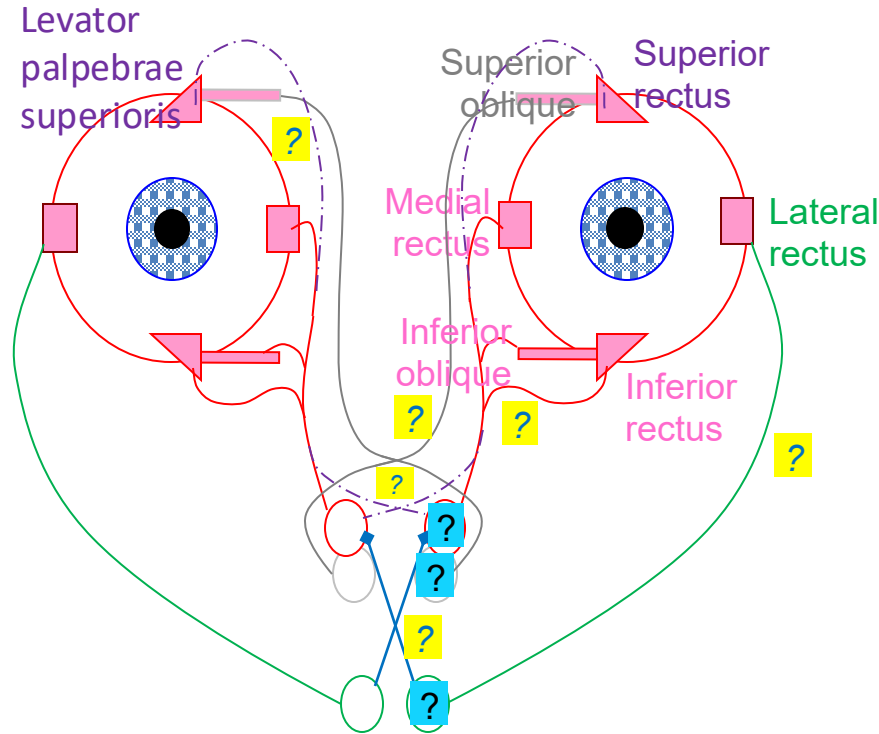
Exceptions (cortico-bulbar):

UMN: Lower face is innervated by contralateral UMN.

LMN: Eye muscle innervations nucleus vs. nerve difference

Eye Movement – CN III, IV, VI

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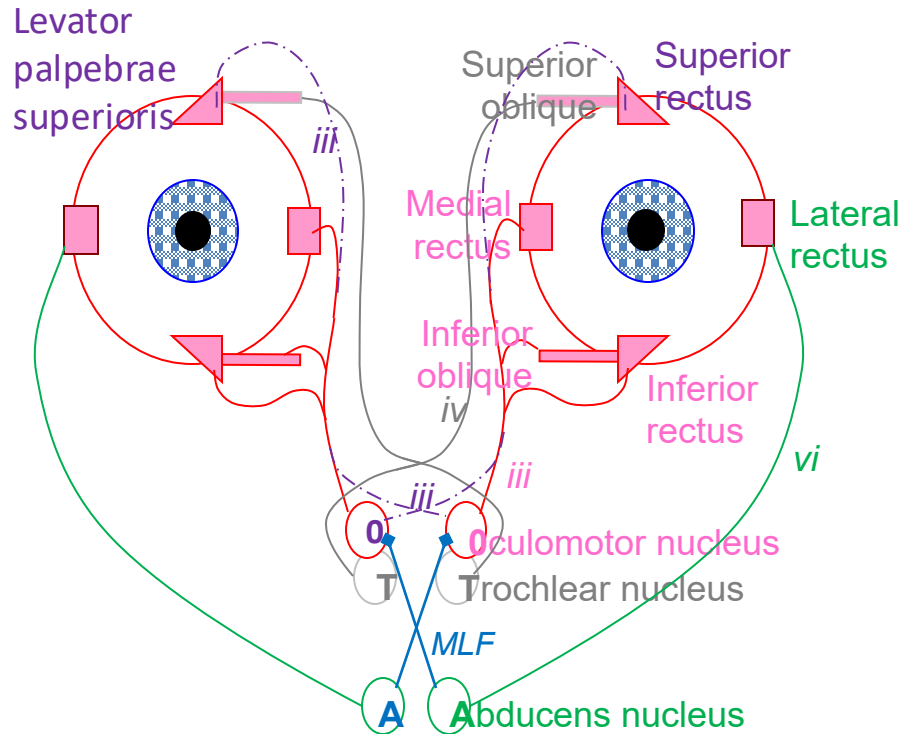
*Which nerve/ fiber
connection? Function?*

Which nucleus?
Function?

Presentation differences of
nerve vs. nucleus damage?

Pause here.....

Eye Movement – CN III, IV, VI



CN III (oculomotor):

- inferior division (MR, IR, IO) + EW nucleus (parasympathetic) → **bilateral** pupil sphincter & ciliary muscles
- superior division (SR & LPS – from **contralateral** oculomotor nucleus)

CN IV (trochlear):

- SO – from **contralateral** trochlear nucleus

CN VI (abducens):

- LR
- Abducens nucleus also activates **contralateral** oculomotor nucleus via MLF (medial longitudinal fasciculus)

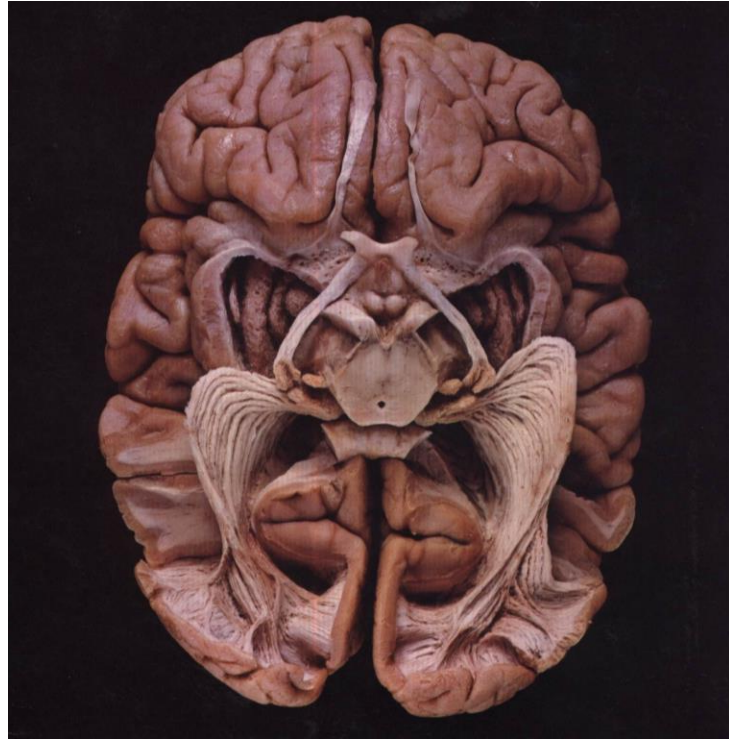
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[illegible]

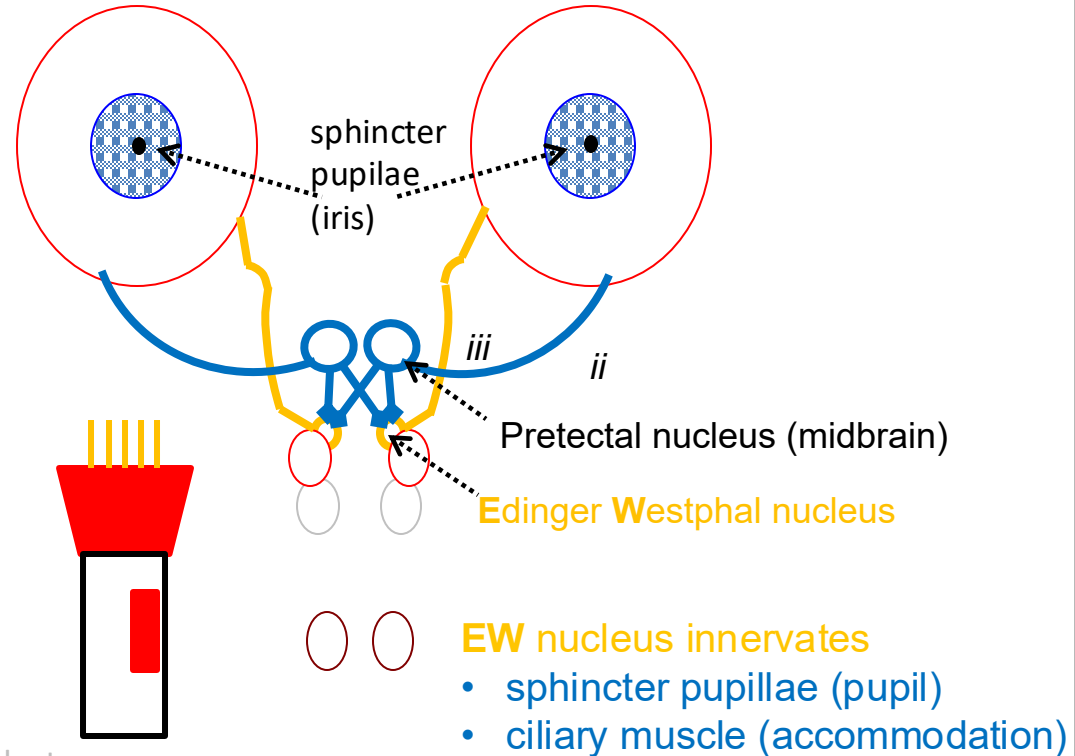
Which nucleus?

Light Reflex

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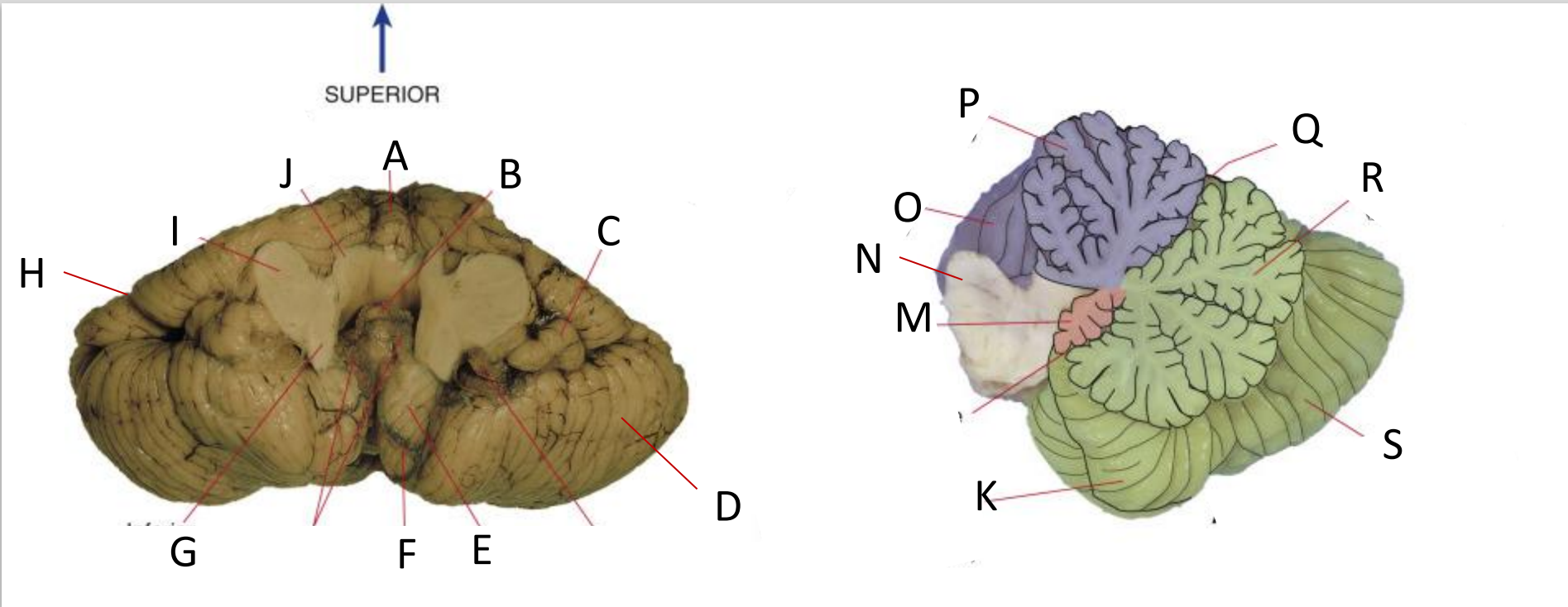


retina → CN II → Pretectal nucleus → Bilateral EW nucleus → each CN III → each iris



Cerebellar Structures

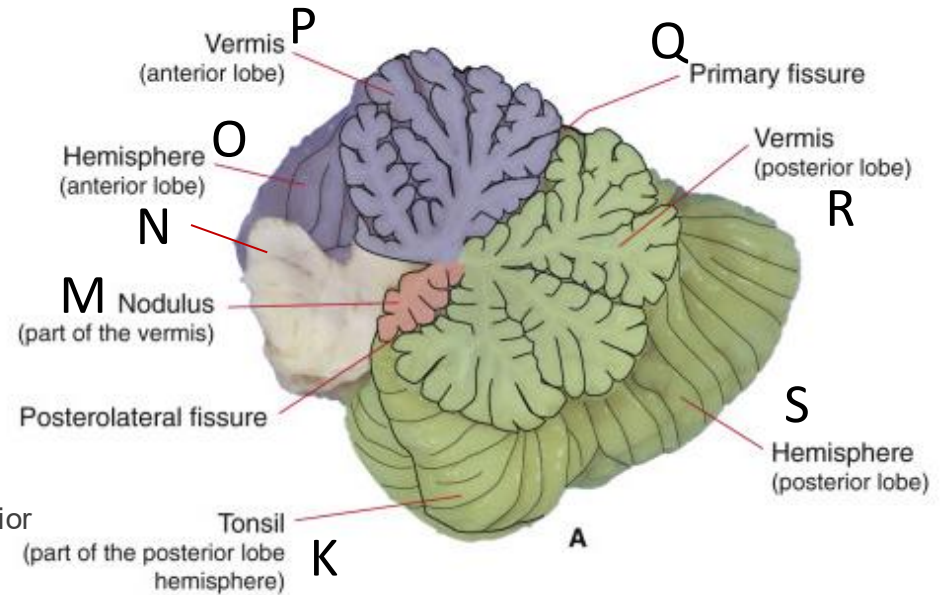
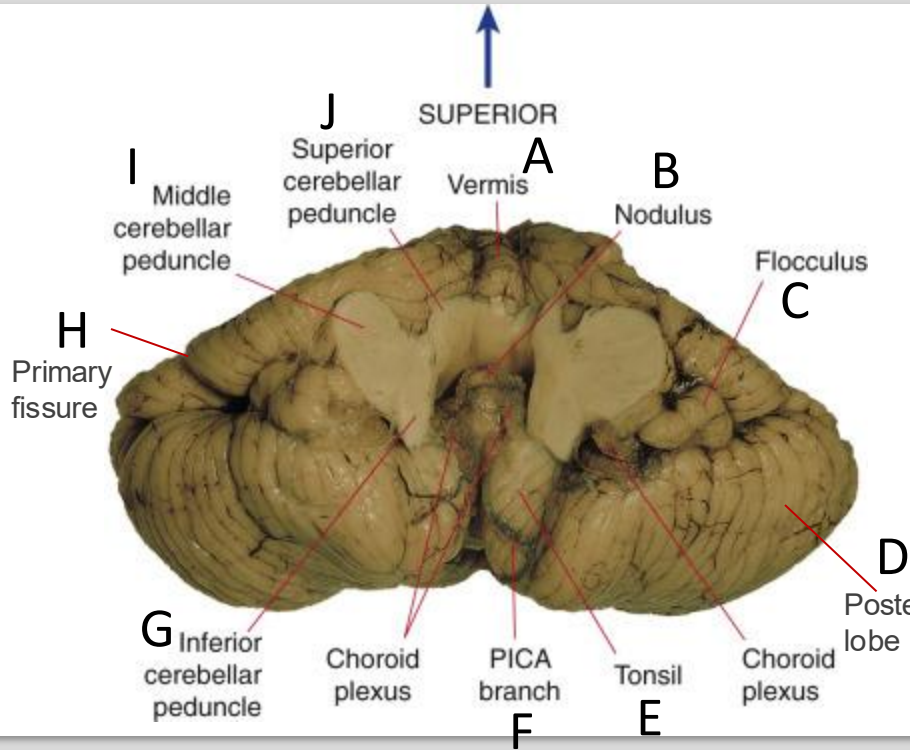
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Pause here.....

Cerebellar Structures

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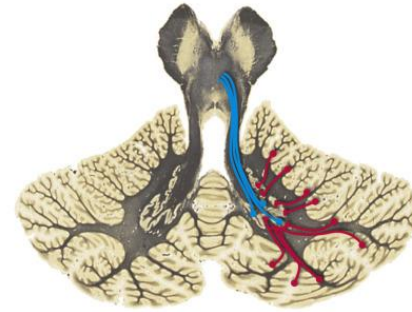
Cerebellar peduncles

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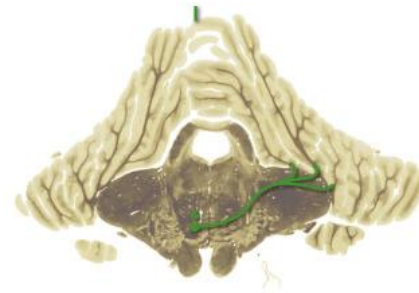
Info from?

Destination?

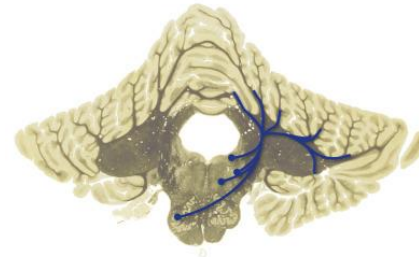
Which peduncle?



Which peduncle?



Which peduncle?

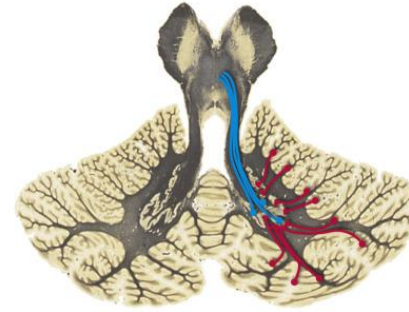


Pause here.....

Info from

Superior CP (SCP)

Cerebellar **effere**nts:
Cortex Purkinje cells → Dentate/
Interposed nuclei → **Ipsi** SCP

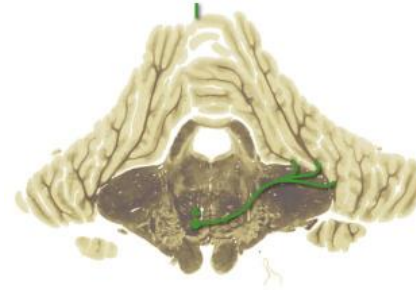


Destination

- **Contra**lateral thalamus (VL) → motor/premotor cortex
- Red nucleus

Middle CP (MCP)

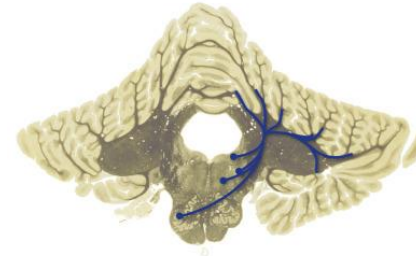
Cerebellar **affer**ents:
Corticopontine tract → Pontine nuclei → **Contra** MCP (mossy fiber)



- Cerebellar cortex (mostly lateral/medial part) → Purkinje cells → Dentate/interposed nuclei

Inferior CP (ICP)

Cerebellar **affer**ents:
1. Vestibular nucleus/spinal cord → Vestibulo-/Spino-cerebellar tracts → **Ipsi** ICP (mossy fiber)
2. Inferior olivary nucleus → **contra** ICP (climbing fiber)



- Cerebellar cortex (vermis/flocculonodular/medial/lateral part) → Purkinje cells → Fastigial/ interposed/ dentate nuclei

Cerebellum Summary

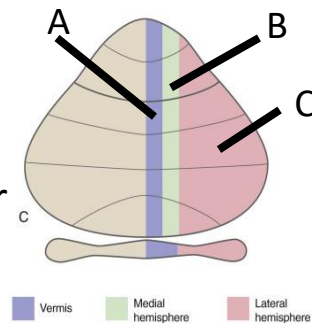
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Input:

- From **spinal cord/vestibular nuclei**: _____ side via _____ cerebellar peduncle at _____ level (smaller portion from the superior cerebellar peduncle)
- From **cerebral cortex**: _____ side via _____ cerebellar peduncle at _____ level

Output:

From **cerebellar cortex** _____ cells → _____ nuclei
(_____, _____, _____ nucleus) → _____ side motor
cortex, via _____ cerebellar peduncle at _____ level



Lesions:

Lateral lesion (lateral cerebellar cortex): _____ ataxia → fall toward _____ side

Medial lesion (vermis, flocculonodular): _____ ataxia (postural instability, broad-based gait) → _____ (side) motor deficits

Damage to the flocculus also affects _____

Cerebellum Summary

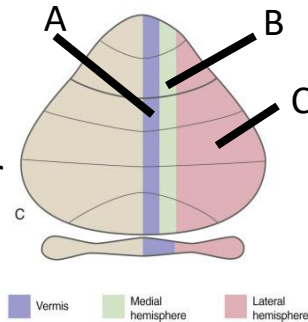
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Input:

- From **spinal cord/vestibular nuclei**: ipsilateral side via inferior cerebellar peduncle at medulla level (smaller portion from the superior cerebellar peduncle)
- From **cerebral cortex**: contralateral side via middle cerebellar peduncle at pons level

Output:

From **cerebellar cortex** Purkinje cells ^{inhibit} → Deep cerebellar nuclei (dentate, interposed, fastigial nucleus) → contralateral side motor cortex, via superior cerebellar peduncle at midbrain level



Lesions:

Lateral lesion (lateral cerebellar cortex): limb ataxia – fall toward ipsilateral side

Medial lesion (vermis, flocculonodular): truncal ataxia (postural instability, broad-based gait) – bilateral motor deficits

Damage to the flocculus also affects eye movements (nystagmus)

Session Objectives

- Review the CNS blood supplies.
- Differentiate the major functions of different lobes.
- Depict the major sensory & motor functional systems.
- Describe the major cerebellar pathways.
- **Integrate the structure – function to identify lesions.**

Case 1

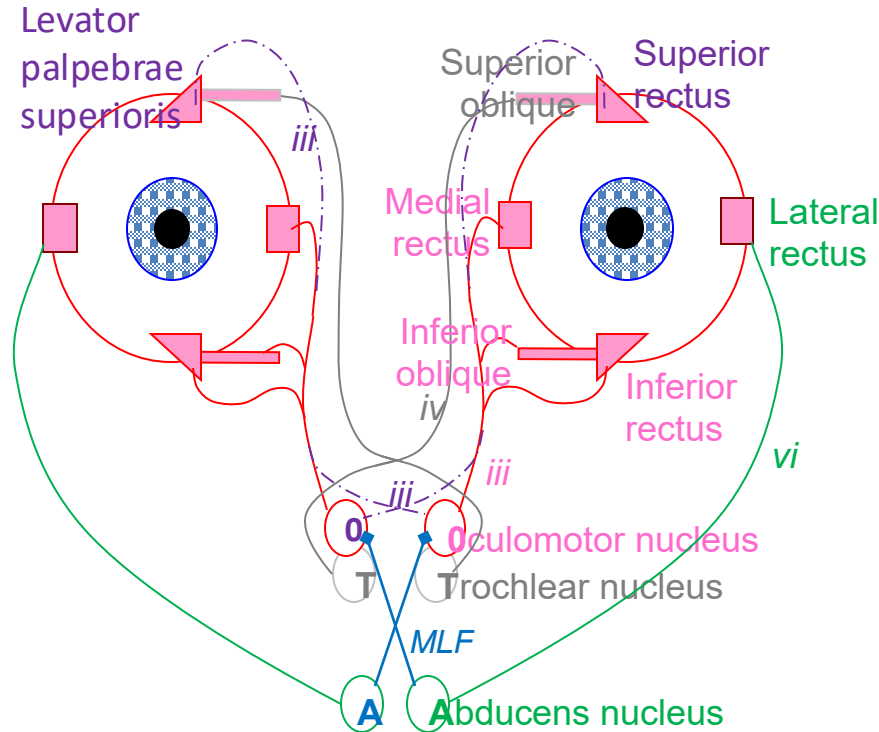
A 35-year-old female presents to the ED with complaints of diplopia and severe headache. History reveals these symptoms have progressed over several days. Physical examination reveals lateral strabismus, ptosis, and pupillary dilation for the left eye. Motor examination reveals normal strength and range of motion of upper and lower extremities. A CT scan of the head and orbits reveals an aneurysm. Which of the following is the most likely location for this aneurysm?

- A) Anterior communicating artery
- B) Left anterior cerebral artery
- C) Left posterior communicating artery
- D) Right posterior cerebral artery
- E) Right posterior communicating artery

Damaged
structure(s)?

Which side?

Eye Movement – CN III, IV, VI



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Which side?

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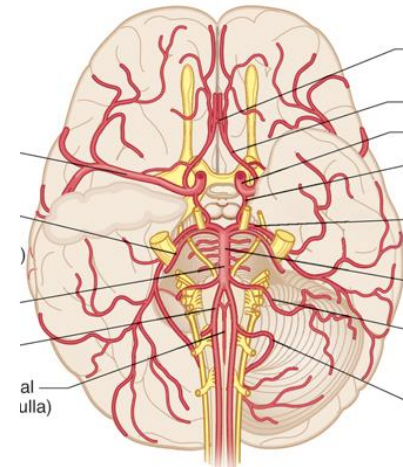
- A) Anterior communicating artery
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Which artery?

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Left
oculomotor
nerve

Case 2

A 25-year-old man comes to the emergency department because he has been dizzy and nauseated and unstable while walking since undergoing a chiropractic treatment for neck pain several hours ago. Physical examination reveals absent pinprick and temperature sensation on his right face, left trunk, and limbs; ataxia of his right extremities; left-beating nystagmus; and hoarseness. Which of the following additional physical findings is most likely in this patient?

- A) Deviation of the tongue to the right
- B) Diminished vibratory sensation in the right extremities
- C) Hyperreflexia of the left extremities
- D) Inability to recognize the left side of his body
- E) Right-sided ptosis and miosis

What structures
were damaged?

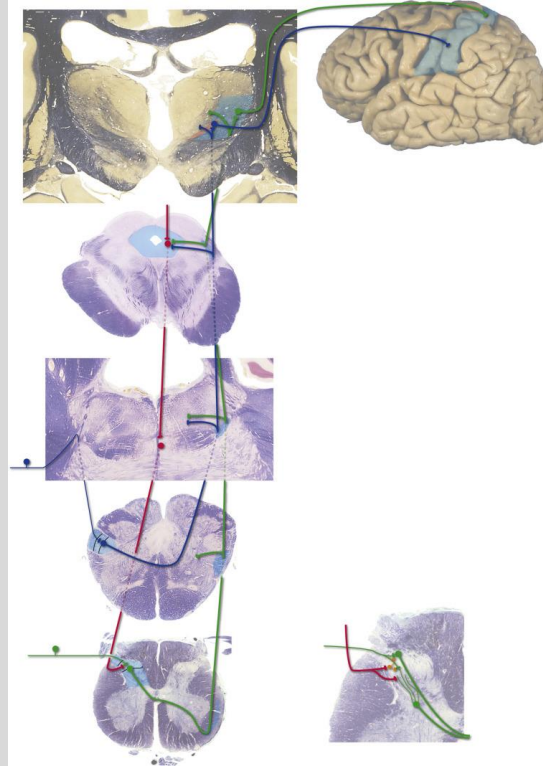
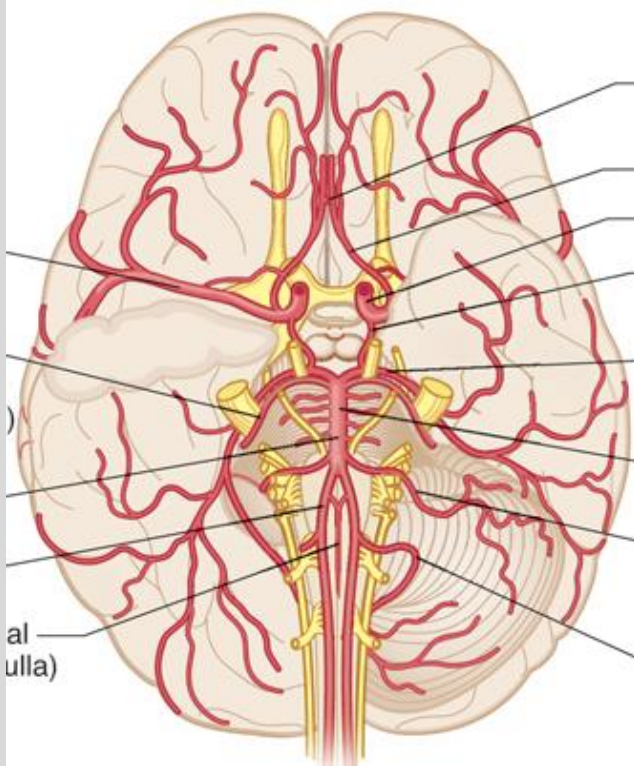
At what level?

Which side?

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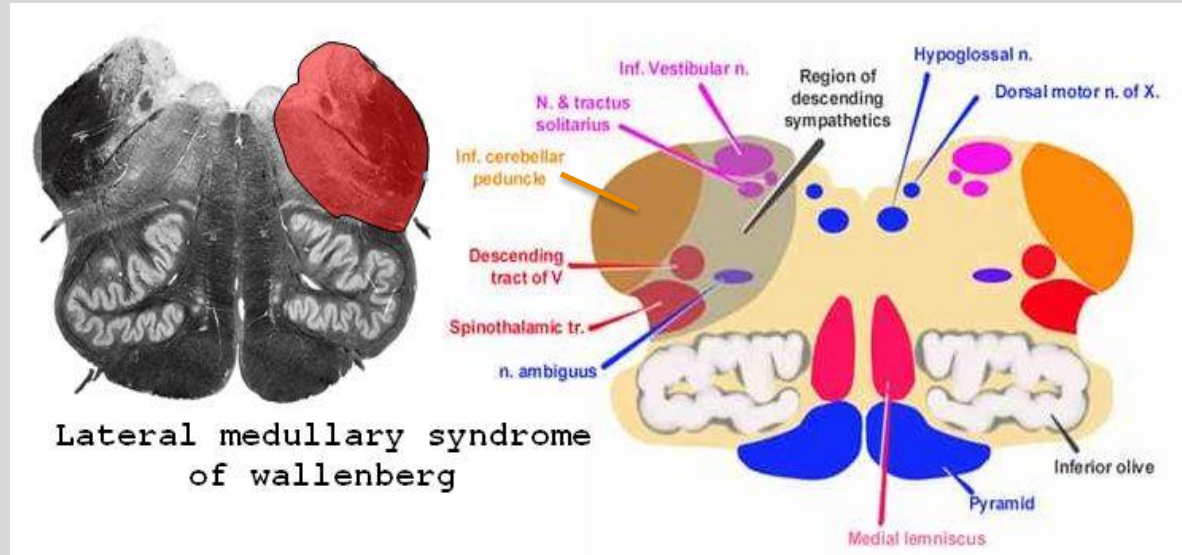


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Horner's syndrome

Lateral medullary syndrome

Which blood vessel stroke is most likely causing it?

PICA or vertebral a. branches